TEMASEK JUNIOR COLLEGE
Preliminary Examinations 2014
Higher 2

BIOLOGY

9648/01

Paper 1 Multiple Choice

Friday, 19 September 2014
1 hour 15 minutes

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name, Centre number and index number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
The use of an approved scientific calculator is expected, where appropriate.

1 Four students, **A**, **B**, **C** and **D**, who observed a root hair cell from a plant using an electron microscope, were asked to confirm the presence or absence within the cell of described cellular structures.

Which student made the correct set of observations?

	Structure with a double membrane inside which are stacks of flattened membranes	Area near the nucleus containing a pair of structures that are composed of microtubules	Structure with a double membrane with inner membrane infolded	Network of tubular-shaped membranous sacs with no ribosomes visible on outer surface of membranes
A	✓	✓	✓	✗
B	✓	✗	✗	✓
C	✗	✓	✓	✓
D	✗	✗	✓	✓

Key
✓ = present
✗ = absent

2 A ribosome consists of a large and a small subunit, each subunit containing ribosomal RNA (rRNA) complexed with a protein.

Which sequence of events concerning ribosomes is correct?

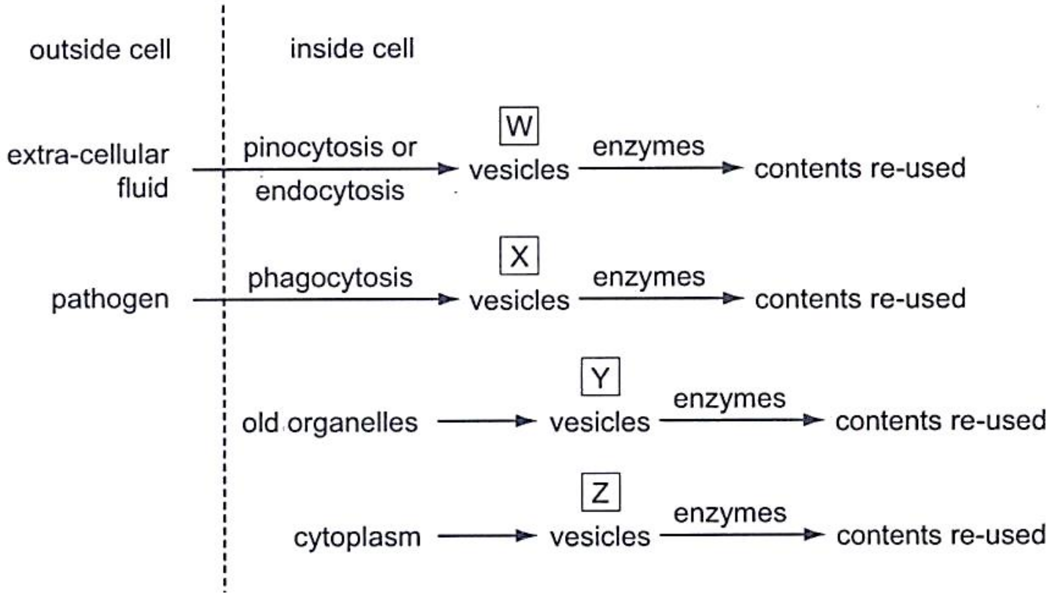
A Within the nucleolus, rRNA and protein are synthesised and subunits formed. They become membrane bound as they are exported through the nuclear envelope to the cytoplasm and RER.

B rRNA and protein are synthesised in the Golgi body and are transported to the nucleolus for subunit formation.

C rRNA is synthesised in the SER and proteins are synthesised by the RER. Subunit formation occurs within the cytoplasm for free ribosomes and within the RER for attached ribosomes.

D rRNA synthesised within the nucleus is complexed with protein that has been imported from the cytoplasm. The subunits formed are exported to the cytoplasm via the nuclear pores.

3 The flow chart shows processes which takes place inside animal cells.



Which processes require the activity of lysosomes?

A **W**, **X**, **Y** and **Z**

B **W** and **X** only

C **X** and **Y** only

D **Y** and **Z** only

4 The table shows the results of an experiment on the rate of uptake of sugars by rat intestine.

sugars		rate of absorption relative to glucose	
		under aerobic conditions	under anaerobic conditions
hexose sugars	glucose	100	30
	galactose	106	32
pentose sugars	xylose	32	32
	arabinose	30	30

Which conclusion may be drawn from the results?

A All four sugars can be absorbed by active transport.

B Only hexose sugars can be metabolised by the rat.

C The intestine is freely permeable to hexose sugars.

D The rate of passive diffusion of all four sugars into the gut is similar.

5 Four students, **A**, **B**, **C** and **D**, were given the same sequence of amino acids removed from a collagen molecule. Each student was asked to analyse the sequence and to explain how their analysis could be linked to a feature of collagen.

Which student's statement shows a correct feature of collagen linked to a correct analysis of the amino acid sequence?

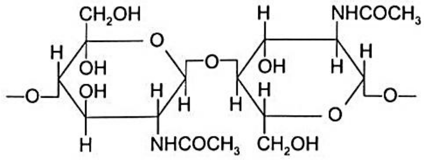
A Collagen has polypeptides arranged parallel to each other and the sequence contains a large variety of amino acids with different sized R-groups.

B Collagen has polypeptides that are arranged very closely together and the sequence has every third amino acid as glycine.

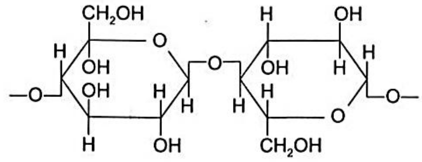
C Collagen has three polypeptides that can fold into a globular structure and the sequence contains cysteine and amino acids with hydrophobic R-groups.

D Collagen is an insoluble molecule and the sequence contains a large proportion of amino acids with hydrophilic R-groups.

6 The diagrams show short sections of some common polysaccharides and modified polysaccharides.



1



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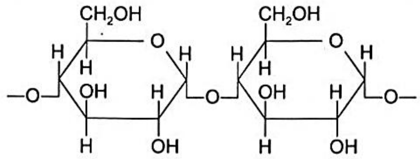

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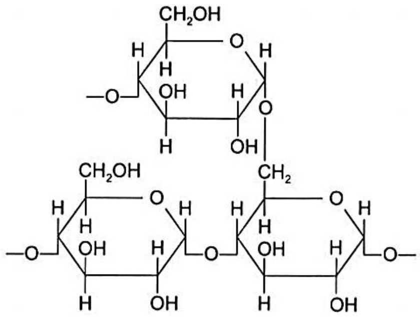

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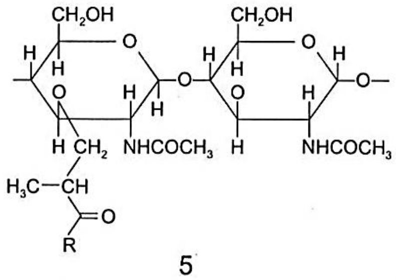


5

The polysaccharides can be described as below.

- polysaccharide **F** is composed of β-glucose monomers with 1,4 glycosidic bonds
- polysaccharide **G** is composed of α-glucose monomers with 1,4 and 1,6 glycosidic bonds
- polysaccharide **H** is composed of N-acetylglucosamine and N-acetylmuramic acid monomers with β-1,4 glycosidic bonds
- polysaccharide **J** is composed of α-glucose monomers with 1,4 glycosidic bonds
- polysaccharide **K** is composed of N-acetylglucosamine monomers with β-1,4 glycosidic bonds

Which shows the correct pairings of polysaccharide descriptions and diagrams?

	polysaccharide **F**	polysaccharide **G**	polysaccharide **H**	polysaccharide **J**	polysaccharide **K**
A	2	4	5	3	1
B	2	5	4	1	3
C	3	4	1	2	5
D	3	5	4	1	2

7 Phospholipids, glycolipids, glycoproteins and proteins are all molecules found in the membranes of cells.

Which type of bond is present in each type of molecule?

Bond types:
1 ester
2 glycosidic
3 ionic
4 peptide

	Phospholipid	Glycolipid	Glycoprotein	Protein
A	1	2	4	3
B	1	3	2	4
C	3	4	1	2
D	3	1	2	4

8 The enzyme rennin is found in gastric juice of young mammals. It causes the clotting of milk protein. The activity of rennin was investigated by recording the time taken for rennin to clot milk in different conditions. The table shows the different conditions used and the results of the investigation.

Tube number	solutions added to 10cm^3 of milk at 35°C						time for milk to clot/min
	2% renin	water	calcium nitrate	calcium citrate	lead nitrate	hydrochloric acid	
1	5 cm^3		2 cm^3			2 cm^3	1
2	5 cm^3				2 cm^3	2 cm^3	no clot formed
3		5 cm^3	2 cm^3			2 cm^3	no clot formed
4	5 cm^3	2 cm^3		2 cm^3			25
5	5 cm^3	2 cm^3				2 cm^3	10
6	5 cm^3			2 cm^3		2 cm^3	2

What is a correct conclusion?

A Calcium ions increase the activity of rennin.

B Citrate ions are necessary for the activity of rennin.

C Hydrochloric acid is necessary for the activity of rennin.

D Nitrate ions inhibit the activity of rennin.

9 The diagram shows a metabolic pathway controlled by end product inhibition.

$$\begin{tikzcd}
 & & \text{metabolite 4} \arrow[r] & \text{metabolite 5} \arrow[r] & \text{end product 6} \\
\text{metabolite 1} \arrow[r] & \text{metabolite 2} \arrow[r] & \text{metabolite 3} \arrow[u] \arrow[d] & & \\
 & & \text{metabolite 7} \arrow[r] & \text{metabolite 8} \arrow[r] & \text{end product 9}
\end{tikzcd}$$

An increase in the concentration of the end product increases above a set level, an enzyme in the pathway leading to the end product is inhibited.

An increase in the concentration of end product 6 does not lead to a decrease in the synthesis of end product 9.

Which enzyme is inhibited by end product 6?

A metabolite 1 → metabolite 2

B metabolite 2 → metabolite 3

C metabolite 3 → metabolite 4

D metabolite 3 → metabolite 7

10 A chemical known to affect mitosis was added, at different stages of mitosis, to actively dividing plant cells with 12 chromosomes.

The results showed that adding this chemical during prophase resulted in cells with 24 chromosomes. Adding the chemical at any other stage resulted in cells with 12 chromosomes.

Which process during mitosis is affected by this chemical?

A Condensing of chromosomes

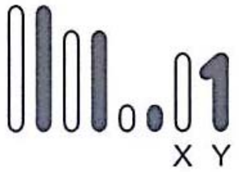
B Organizing the spindle

C Producing the centromeres

D Separating centrioles

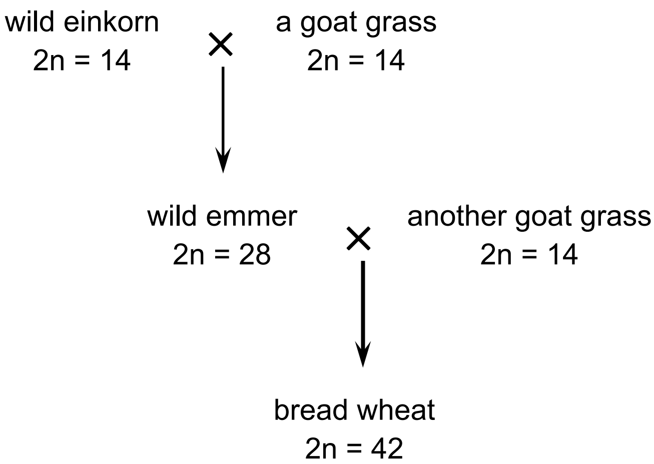
11 No crossing over occurs during meiosis in male fruit flies of the species *Drosophila melanogaster.*

The diagram shows the four pairs of homologous chromosomes present in a testis cell of a male fly.



Which set of chromosomes in a gamete nucleus shows the genetic variation resulting from independent assortment?

12 Bread wheat is the result of crosses between ancient wheat species and goat grasses which have occurred in several stages over the last ten thousand years.



Why does this type of chromosome change **not** normally happen?

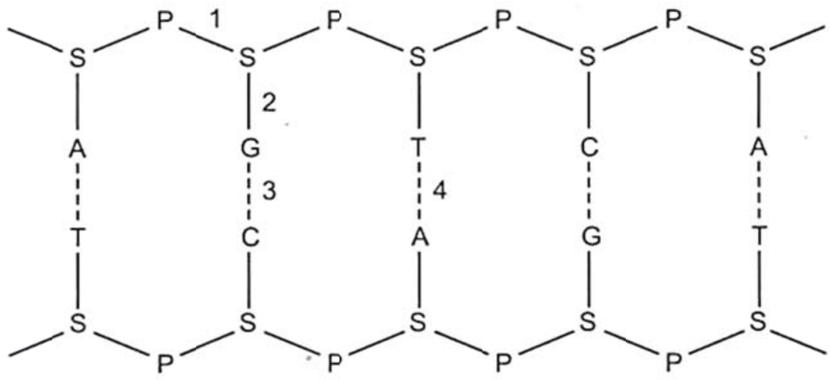
A Diploid cells are formed by mitosis.

B Diploid gametes undergo a reduction division.

C Meiosis halves the diploid number of chromosomes.

D Meiosis results in diploid gametes.

13 The diagram shows part of a nucleic acid.



key
P = phosphate
S = sugar molecule

Which row correctly describes the bonds shown in the diagram at positions 1, 2, 3 and 4?

	Is formed by condensation	Forms a di-ester	Occurs during transcription	Involves attraction between polar molecules
A	1	2	1,2 and 3	3
B	1 and 2	1	3 and 4	3 and 4
C	2,3 and 4	1	3 and 4	1,3 and 4
D	2	3	1,3 and 4	4

14 The structure of a DNA molecule has the features shown below.

1 Four different bases
2 Antiparallel strands
3 Pentose sugars
4 Phosphate groups
5 Supercoiling

Which features make DNA suitable for storing information?

A 1, 2 and 5

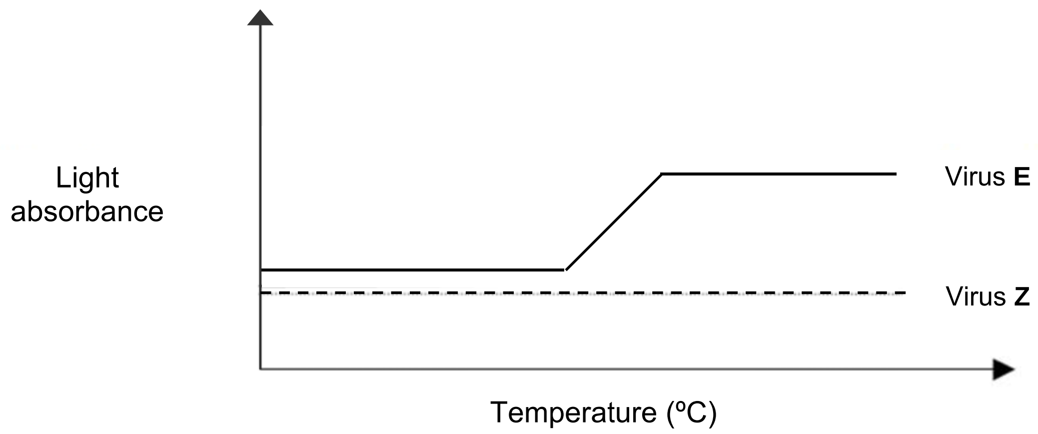
B 1, 4 and 5

C 2, 3 and 4

D 3, 4 and 5

15 Two new viruses **E** and **Z**, which infect eukaryotic cells, have been identified.

In one experiment, the nucleic acid from each virus is isolated and analyzed over a range of temperatures. The light absorbance of nucleic acids changes when denaturation or annealing occurs. The behaviour of the nucleic acid from each virus is shown in the graph.



In a second experiment, it is found that treatment with reverse transcriptase inhibitors or with inhibitors of DNA synthesis blocks the ability of virus **Z** to infect cells. In contrast, reverse transcriptase inhibitors have no effect on the ability of virus **E** to infect cells but DNA synthesis inhibitors block infection by virus **E**.

Which of the following conclusions can be drawn from both the experiments?

A The genome of virus **E** is single-stranded RNA and that of virus **Z** is double-stranded DNA.

B The genome of virus **E** is double-stranded DNA and that of virus **Z** is single-stranded RNA.

C The genome of virus **E** is double-stranded RNA and that of virus **Z** is single-stranded DNA.

D The genome of virus **E** is double-stranded DNA and that of virus **Z** is double-stranded RNA.

16 Papillomaviruses are a diverse group of non-enveloped DNA viruses that infect the skin and mucous membranes of humans and a variety of animals. Scientists have developed a vaccine, which comprises the viral capsid protein, against human papillomaviruses (HPV).

Which stage of the viral replication is directly affected after an individual is vaccinated against HPV?

A attachment stage

B viral synthesis stage

C assembly stage

D release stage

17 The table shows a comparison of some aspects of the genomes and protein-coding genes of the prokaryote bacterium, *Escherichia coli* and the eukaryote fungus *Saccharomyces cerevisiae.*

	E.coli	*S.cerevisiae*
Genome length/base pairs	4 640 000	12 068 000
Number of protein-coding genes	4300	5800
Proteins with roles in:		
Metabolism	650	650
Energy release/storage	240	175
Membrane transport	280	250
Transcription	240	400
Translation	180	350
Cell structure	180	250

What could **not** account for the differences in the number of protein-coding genes?

A Many catabolic pathways for using carbon compounds in prokaryotes.

B The presence of introns in the DNA of eukaryotes.

C The presence of membrane-bound organelles in eukaryotes.

D The use of histones to package DNA in eukaryotes.

18 The following are characteristics of eukaryote transcription.

- Promoters are activated by transcription factors that recognise specific DNA sequences and other sequences that are very similar.
- Within a promoter, there may be recognition sites for more than one transcription factor.
- Similar specific DNA sequences can be recognised by more than one transcription factor.
- Each transcription factor may be capable of recognising a number of promoter recognition sites.

What explains the different levels of expression of a eukaryotic gene?

A Competition between recognition sites present in the promoter for transcription factors.

B Competition between transcription factors that recognise the same sites of a promoter.

C The number of transcription factors that recognise the same sites of a promoter.

D The number of types of different transcription factors.

19 The processing of pre-mRNA involves the following steps to produce functional mRNA.

1 polyadenylation
2 removal of introns
3 splicing of exons
4 capping

In which order do they occur?

A $1 \rightarrow 3 \rightarrow 2 \rightarrow 4$

B $2 \rightarrow 4 \rightarrow 1 \rightarrow 3$

C $3 \rightarrow 1 \rightarrow 4 \rightarrow 2$

D $4 \rightarrow 2 \rightarrow 3 \rightarrow 1$

20 The human papillomavirus (HPV) may transform its target cell into a cervical cancer cell, which can lead to a tumour.

Which statement correctly explains this transformation?

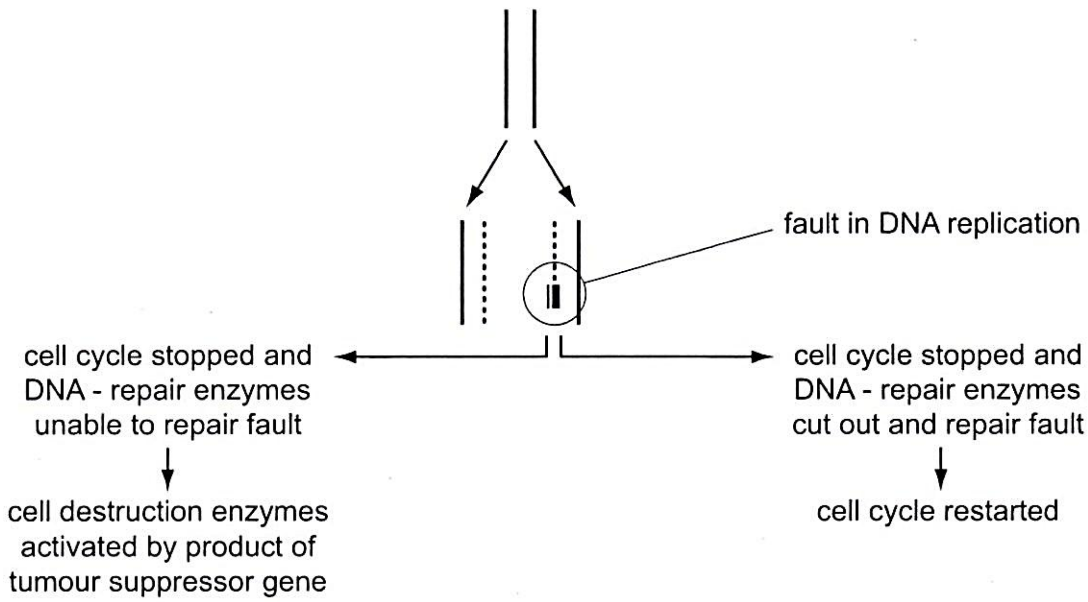
A An oncogene codes for a protein that causes cell death.

B An oncogene codes for a protein that results in the cell dividing indefinitely.

C A proto-oncogene codes for a protein that causes cell death.

D A proto-oncogene codes for a protein that slows cell division.

21 The diagram shows the possible consequences of a fault in DNA replication.



Which statement explains why mutations in the tumour suppressor gene may lead to the development of cancer?

A Faulty DNA will not be recognized by the tumour suppressor gene.

B Programmed cell death will not be activated by the tumour suppressor gene.

C The cell cycle will not be restarted by the tumour suppressor gene.

D The cell cycle will not be stopped by the tumour suppressor gene.

22 The speech defect known as stuttering may involve two genes, G and N. Most people who are homozygous for the alleles g and n are not stutterers.

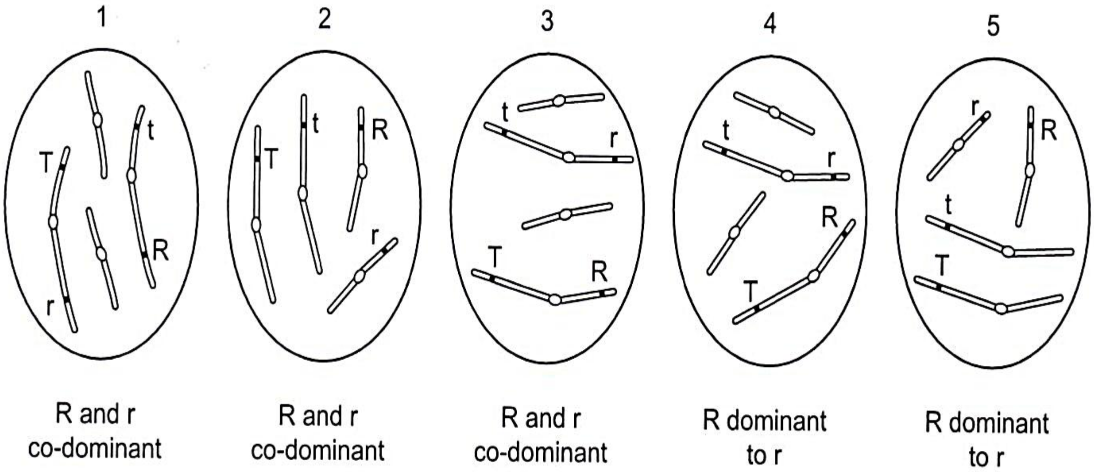
However, recent research has shown that the presence of either of the mutant alleles G or N can cause stuttering in heterozygotes.

Using this information, which proportion of the children of a couple, the father with genotype Ggnn and the mother ggNn, are likely to be stutterers?

A 3/16 **B** 6/16 **C** 9/16 **D** 12/16

23 In a family of flowering plants, height is controlled by a pair of alleles. The allele for tall (T) is always dominant to the allele for short (t). The flower colour is also controlled by a pair of alleles. In some species, the allele for red (R) is dominant to the allele for white (r). In other species the colour alleles are co-dominant. (For simplicity, the symbols R and r are used for the co-dominant alleles.)

The diagram shows the chromosome arrangement and information about the height alleles and the flower colour allele in five species of this family of plants.



Each of the plants 1, 2, 3, 4 and 5 was test crossed.

Assuming there is no crossing over, which plants would produce offspring with the phenotypes short with white flowers and tall with red flowers?

A 1, 3 and 4

B 2, 4 and 5

C 2 and 5 only

D 4 and 5 only

24 Which statement concerning chrysanthemum plants, of the genus *Dendranthema,* is a valid example of how the environment may affect the phenotype?

A Anthocyanins and anthoxanthins are vacuolar pigments, whereas xanthophylls and carotenes are pigments found in membrane-bound organelles known as plastids. These, together with molecules known as co-pigments, are responsible for the variation observed in petal colour in *Dendranthema.*

B Identical genetic crosses performed between varieties of *Dendrathema* result in a greater proportion of offspring plants with plastids exhibiting a yellow colour when grown in a field and a greater proportion of offspring plants with colourless plastids when grown in a glasshouse.

C The seeds of a cross between *Dendranthema weyrichii* and *Dendranthema grandiflora* produce plants that are far more frost-tolerant and exhibit an extended flowering season compared with both parent plants.

D The seeds of a cross between *Dendranthema weyrichii* (height varying between 12.5 – 15.0 cm) and *Dendranthema grandiflora* (height varying between 8.0 – 25.0 cm) produce plants, when grown in natural day length, of a height varying between 55.0 – 71.0 cm.

25 Which of the following statements are true for both cyclic and non-cyclic phosphorylation?

I Hydrolysis of ATP occurs in both cyclic and non-cyclic phosphorylation.

II Electron displacement occurs in the reaction centre of photosystem I in both cyclic and non-cyclic phosphorylation.

III Energy released from the electron transport chain is used to pump protons from the stroma into the thylakoid lumen.

IV $NADP^+$ is oxidized in both cyclic phosphorylation and non-cyclic phosphorylation.

A **I** and **IV**

B **II** and **III**

C **II** and **IV**

D **II**, **III** and **IV**

26 Which of the following processes do **not** occur during the conversion of glucose to two molecules of pyruvate?

I hydrolysis of ATP

II phosphorylation of hexose

III release of CO_2

VI reduction of NAD

V oxidative phosphorylation of ADP

A **I** and **III** only

B **II** and **IV** only

C **III** and **V** only

D **I**, **II** and **V** only

27 Which of the following statements are true?

I Receptor tyrosine kinases are able to phosphorylate tyrosine residues.

II All receptor tyrosine kinases comprise of 2 subunits which come together upon ligand binding.

III G-protein coupled receptors have GTPase enzymes which hydrolyze the G-protein bound GTP molecule to GDP**.**

IV Hydrophobic ligand molecules typically bind to intracellular receptors.

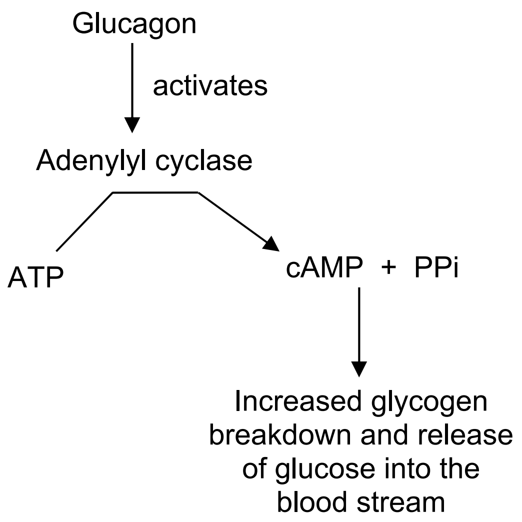
A **I** and **IV**

B **II** and **IV**

C **II**, **III** and **IV**

D All the above statements are true

28 One action of the hormone glucagon is to cause the activation of adenylyl cyclase in liver cells. Adenylyl cyclase is an enzyme that converts ATP into the messenger molecule cyclic AMP (cAMP), the presence of which triggers a cascade of events within the cell that leads to the breakdown of glycogen and the subsequent release of glucose into the blood stream. This pathway is summarised below.



When glucagon is removed, adenylyl cyclase becomes inactive again and ceases to convert ATP to cAMP. The enzyme phosphodiesterase converts cAMP into AMP. The following table summarises an experiment measuring glucose release by liver cell cultures under various conditions.

culture	condition(s)	Observation(s)
1	no addition of glucagon and caffine	minimal glucose release
2	glucagon added and later removed	glucose released into media; rapid reduction in release after glucagon removed
3	caffeine added	minimal glucose release
4	glucagon and caffeine added; glucagon was then removed but caffeine maintained	glucose released into media; slow reduction in release after glucagon removed

Based on these observations, which of the following statement is most likely to be true?

A Caffeine inhibits the action of phosphodiesterase.

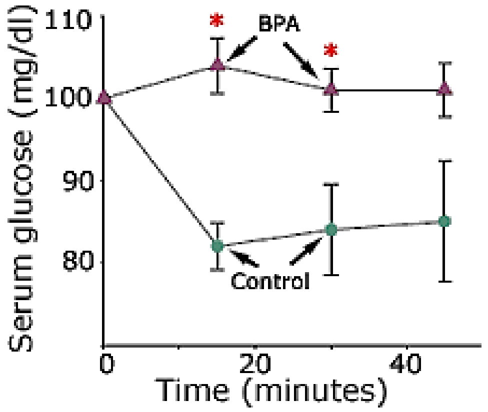
B Caffeine stimulates synthesis of cAMP.

C Caffeine binds the active site of adenylyl cyclase.

D Caffeine prevents glucagon breakdown.

29 An experiment was performed on laboratory mice to test the effects of bisphenol A (BPA), a chemical commonly used to make polycarbonate bottles. BPA was fed orally to the mice and their serum glucose levels were tested at various time intervals.

The graph shows the results of the experiment.



Based on the results shown, which of the following **cannot** be a long-term effect of BPA?

A BPA-treated animals will secrete more insulin than the control.

B Functions of β cells of the islets of Langerhans will be affected.

C Hypoglycaemia will develop with long term use of BPA.

D Insulin resistance will develop.

30 Which feature of the transmission of nerve impulses does **not** have a role in maintaining the unidirectional movement of that signal?

A Exocytosis of Golgi vesicles at the presynaptic membrane.

B Neurotransmitter-hydrolysing enzyme located on the postsynaptic membrane.

C Neurotransmitter receptors located on the postsynaptic membrane.

D Sodium-gated channels on the postsynaptic membrane.

31 Four events in the transmission of nerve impulses across synapses are:

1 depolarisation of the presynaptic membrane
2 propagation of postsynaptic action potential
3 hydrolysis of transmitter substance
4 rupturing of synaptic vesicles

In which sequence do these events occur?

	first	→		*last*
A	1	3	2	4
B	1	4	2	3
C	4	1	3	2
D	4	3	1	2

32 Many types of evidence, including homology, can provide evidence in support of Darwin's theory of natural selection.

Which statement does **not** provide support?

A The allele for sickle cell haemoglobin that gives resistance to malaria is more frequent in malarial areas.

B The distribution of variants of the A blood group antigen reflects human migration patterns.

C The homozygous condition of the sex-linked allele for a non-functional blood clotting protein is rare.

D The molecular structure of ATP is almost identical in all eukaryotes.

33 It is thought that the original human population of North America descended from as few as twenty individuals moving across the land bridge from Asia at the end of the last Ice Age.

Why is the degree of homozygosity higher in their descendents than in this original population?

A directional selection

B founder effect

C genetic drift

D reduced mutation

34 The pigment haemoglobin found in red blood cells of mammals and birds combines readily with oxygen.

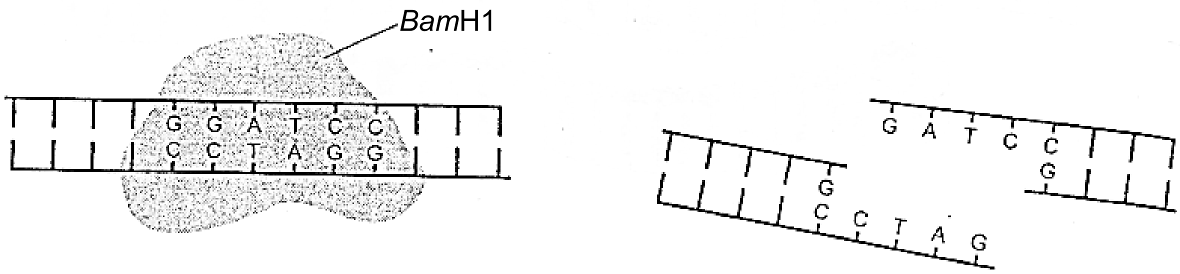
DNA analysis has revealed that a form of haemoglobin is found in a wide range of unrelated phylogenetic groups including bacteria, annelids, anthropods and leguminous plants.

Which evolutionary processes could account for the distribution of haemoglobin in such a wide variety of organisms?

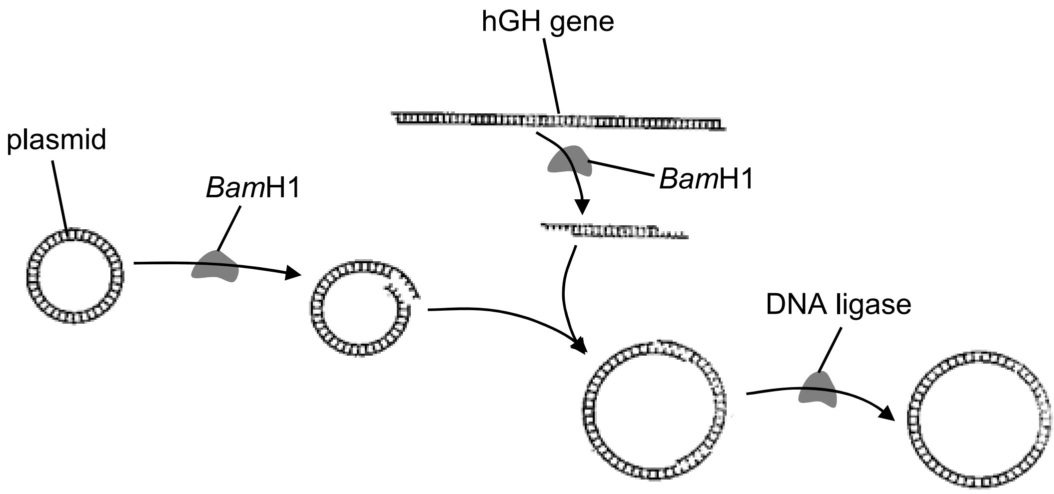
	Adaptive radiation	Conservation of genes	Natural selection
A	✓	✓	✓
B	✓	✓	✗
C	✓	✗	✓
D	✗	✓	✓

Key
✓ = present
✗ = absent

35 *Bam*H1 is a restriction enzyme that cuts DNA as shown in the diagram.



The diagram below shows part of the procedure for producing *E.coli* that will synthesise human growth hormone, hGH.



At the end of this process, many plasmids do not contain the hGH gene.

What could explain this?

A Different alleles of the hGH gene have different sticky ends.

B Not all of the plasmids cut by *Bam*H1 have sticky ends.

C Some of the plasmids are cut at more than one position.

D The sticky ends of some of the plasmids rejoin with each other.

36 A particular restriction enzyme cuts within a specific sequence of three base pairs in DNA.

What explain(s) the absence of this triplet in the genome of the bacterium from which the enzyme was taken?

1 To prevent the enzyme from cutting the bacterium's own DNA
2 So that all DNA containing that triplet is recognized as foreign
3 Because the bacterium does not use the amino acid for which the triplet code

A 1 only

B 1 and 2

C 2 and 3

D 1, 2 and 3

37 One method of making the human form of insulin uses genetically-modified *Escherichia coli.* Artificially synthesized strands of DNA which code for the insulin chains A and B and for the carrier proteins specific to chain A and chain B are spliced into bacterial plasmids. The stages are summarized in the flow chart.

V		**W**		**X**		**Y**
Plasmids opened	→	Synthetic DNA for A and for B spliced into separate plasmids	→	Plasmids closed	→	Plasmids replaced into bacteria

The genetically-modified bacteria are grown in separate fermenters in a medium with the specific antibiotic. The two chains with their carrier proteins are synthesized inside the bacteria. Then:

Z		
Bacteria are broken open	→	Carrier proteins are detached

Which agents are used at stages **V**, **W**, **X**, **Y** and **Z**?

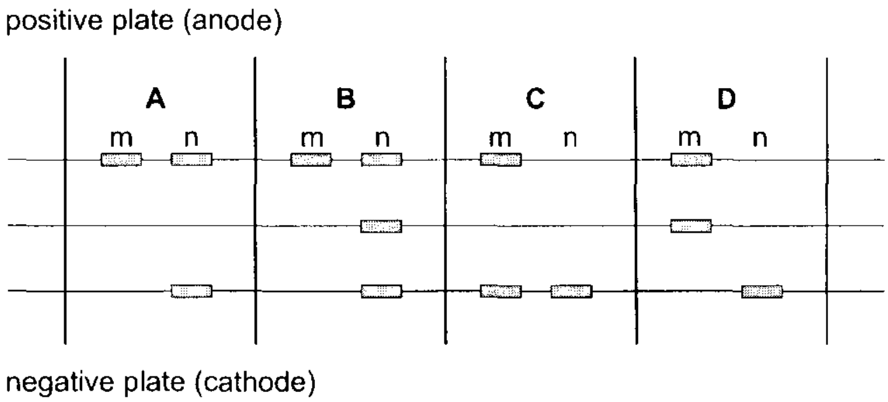
	DNA ligase	Heat shock	Lysozyme	Restriction enzyme
A	**V**	**W**	**X** and **Y**	**Z**
B	**W** and **X**	**Y**	**Z**	**V**
C	**Y**	**Z** and **V**	**W**	**X**
D	**Z**	**X**	**V**	**W** and **Y**

38 There are two forms of the seeds of the garden pea, *Pisum sativum*, one with a wrinkled skin and the other with a smooth skin.

Plants with wrinkled seeds have a recessive mutation caused by the insertion of a transposable element into a gene.

Two heterozygous pea plants were crossed. Two of the offspring (m and n) had the DNA for this gene removed and separated by electrophoresis.

Which pattern of bands shows a wrinkled phenotype and a smooth phenotype?



39 Which uses of the information from the human genome project are generally considered to be unethical?

1 an insurance company only giving cheap rates to people with genetic predispositions to fewer diseases

2 genetic archaeologists identifying the earliest forms of genes to show evolutionary relationships

3 cytologists developing tests for only some defective genes

4 doctors only giving specific drugs to block the actions of faulty genes to carriers of those genes

5 genetic councillors giving specific lifestyle information only to people genetically predisposed to risks

6 parents choosing embryos for implantation only after ante-natal tests for acceptable genes

A 1 and 3 **B** 1 and 6 **C** 2 and 5 **D** 3 and 4

40 Developing fish eggs can be treated to produce a diploid egg. In salmon, such eggs have been fused with haploid salmon sperm to give infertile triploid salmon.

Reproductive organ tissue from diploid trout was transplanted into newly hatched triploid salmon. This tissue matured as the fish grew and the salmon successfully produced viable trout sperm or eggs which resulted in young trout.

Which fish could be seen as genetically modified?

A salmon providing eggs for treatment

B trout providing reproductive organ tissue

C young triploid salmon

D young trout